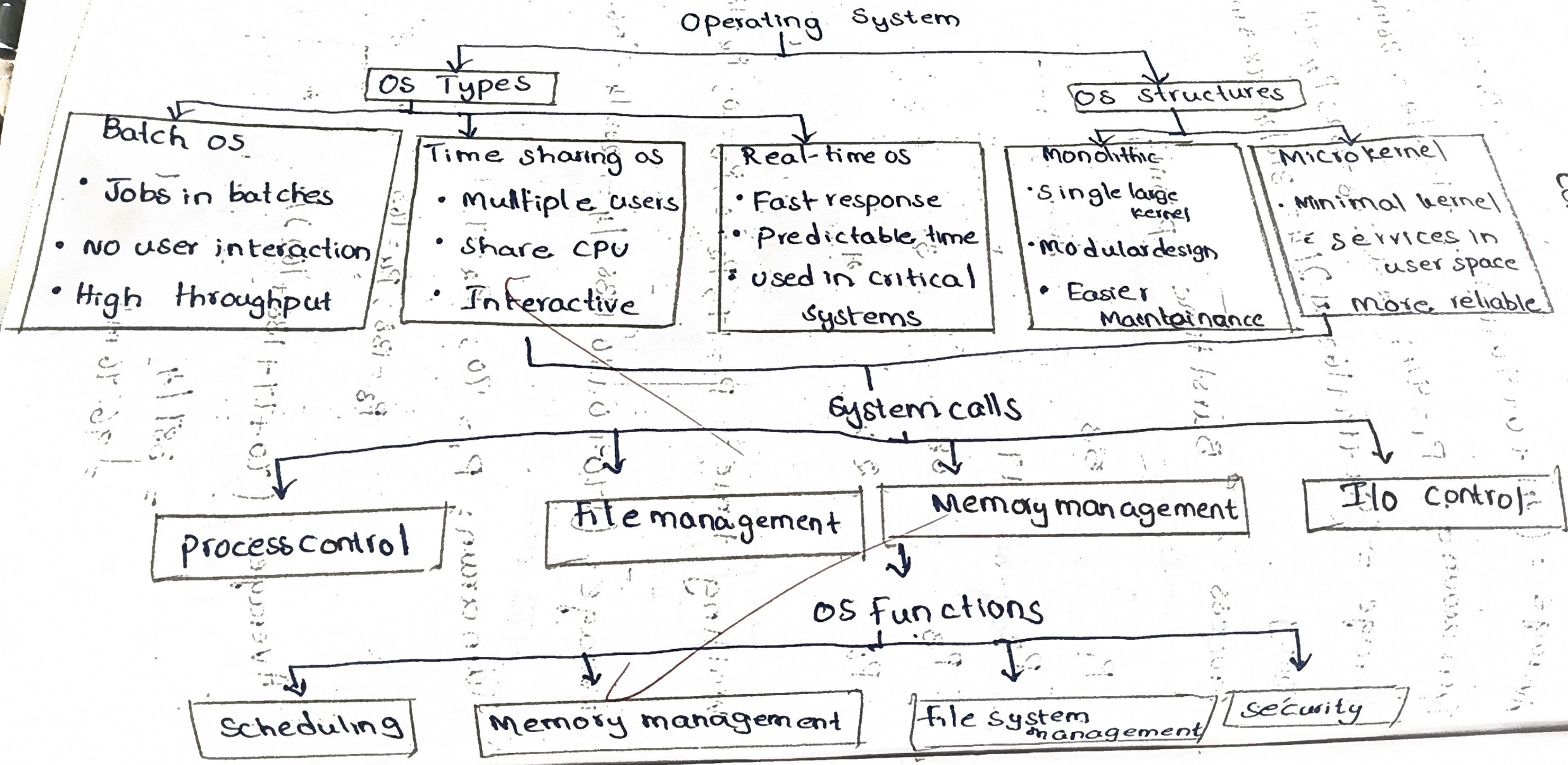
OS Types → OS structures → System calls → OS functions



Concept map

Process, Memory, File and I/O system

OS functions

Scheduling | Memory management | File system management | Security

Process, Memory, File and I/O system

Operating System

Process management
- Create/Terminate
- Scheduling
- Synchronization

Memory management
- Allocation
- Deallocation
- Protection

File system
- open/Read/write
- close/Delete
- File organization

I/O system
- Device Drivers
- I/O control
- Buffering

System calls

Connects user programs with Process + Memory + File + I/O

Interdependencies

Process Management ↔ MemoryManagement

Process management ↔ File systeme ↓ system calls

Filesystem ↔ I/O system

Memory management → I/O system

Comparative Concept Map – Batch, Time-sharing, Distributed & Real-time OS

Operating system

Batch OS

Key Idea
- Jobs processed in batches
- No user interaction
- High throughput

Advantages
- Good for large jobs
- Simple
- Efficient

Disadvantages
- Slow response time
- No interaction

Time-sharing OS

Key idea
- Many users share CPU
- Interactive system
- Quick response

Advantages
- Interactive
- Fast response
- Better resource use

Desadvantages
- overhead
- security issues
- Complex design

Distributed OS

Key idea
- Multiple computers connected
- Resource sharing
- Network dependent

Advantages
- High reliability
- Scalable

Disadvantages
- Network dependency
- More complex

Real-time OS

Key idea
- Fast and predictable response
- used in critical systems

Advantages
- Deterministic response
- Reliable
- Good for time-critical

Disadvantages
- Expensive
- limited resources

Hierarchical + Functional system call Map

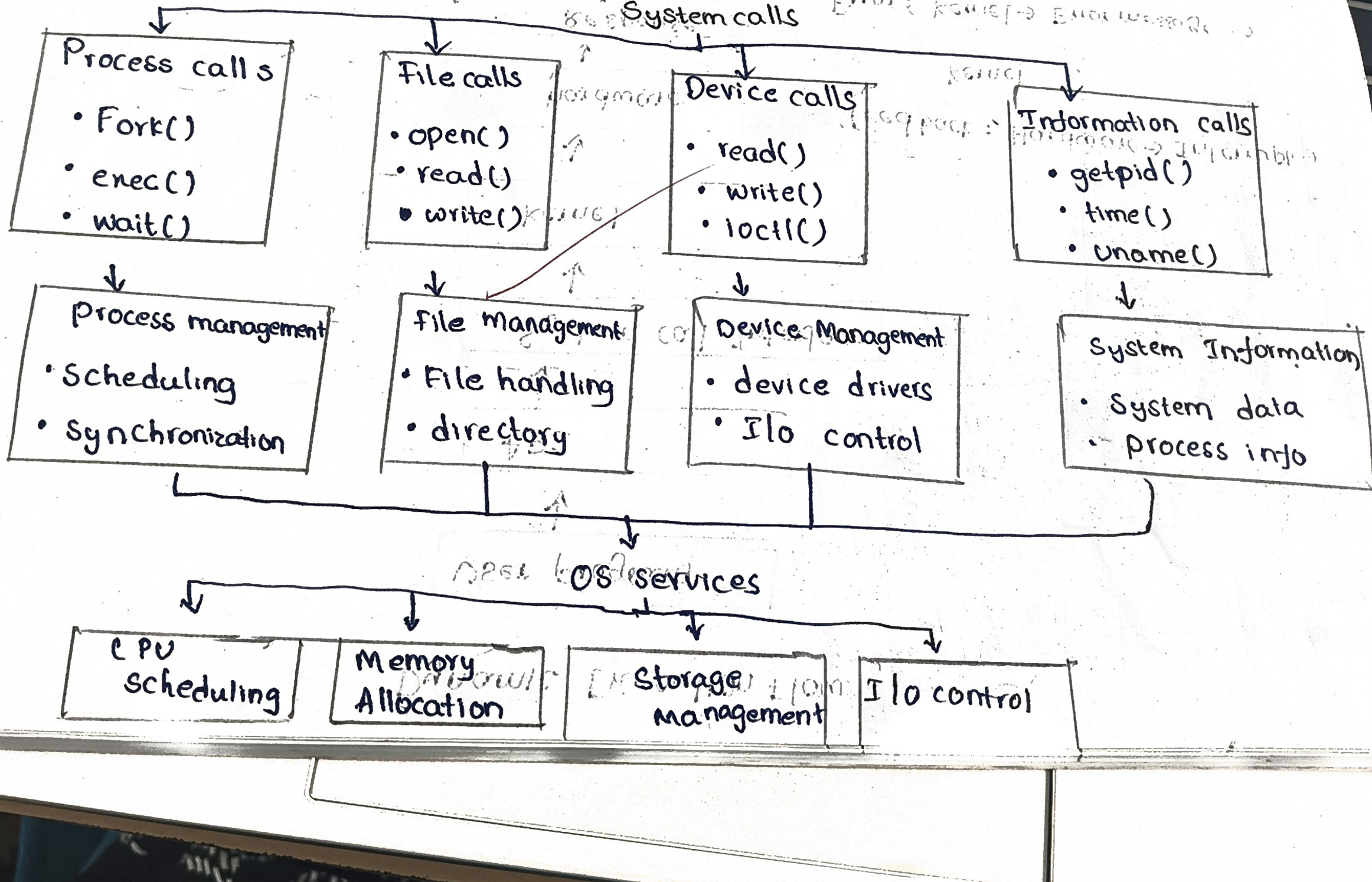
Disadvantages
- Slow response time
- No interaction

- overhead
- security issues
- Complen design

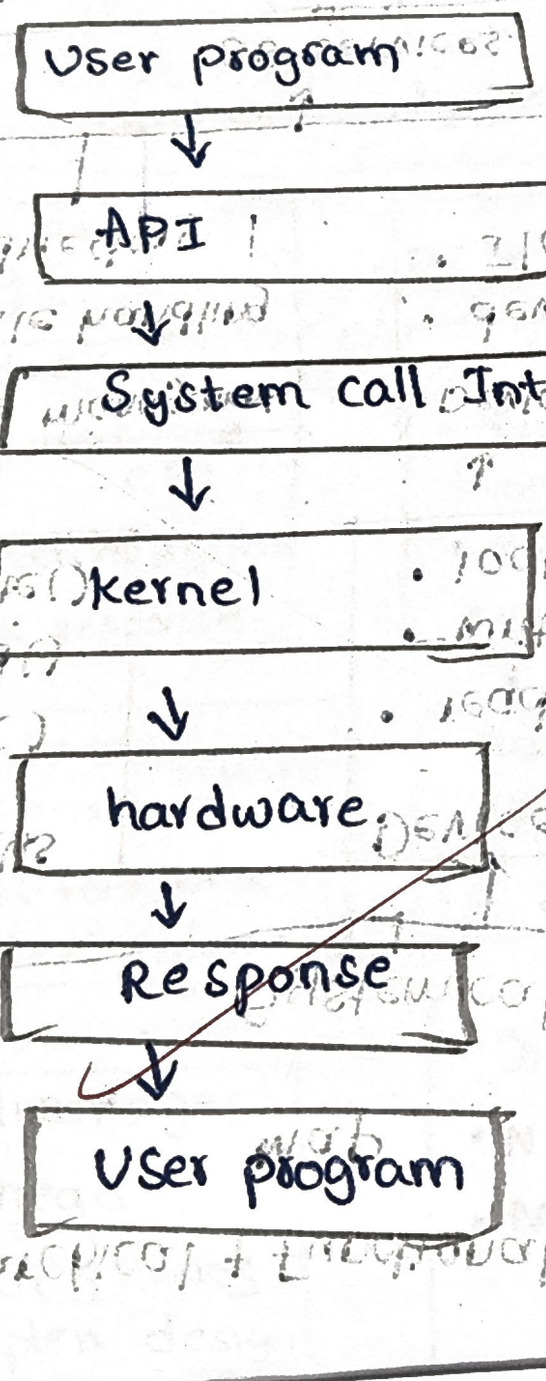
Disadvantages
- Network dependency
- More complen

Disadvantages
- Enpensive
- limited resources

Hierarchical + Functional system call Map



Dynamic Execution Flow



Feedback : Hardware → Interrupt → Kernel

Error : Kernel → Error message → User program